CIP-A106 Laboratory 3

TeslaCrypt Ransomware Unpacking and Memory Analysis

demo1_ransomware.bin: packed 32-bit Windows executable (name of the exercise archive is provenance only)

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Course	AIVTIC CIP-A106, Laboratory 3 (Technical Report)
Analysis period	2026-10-03 11:00 to 2026-10-04 14:00 (VM local time). First, uncontrolled execution at 2026-10-03 12:45:25.
Package	CIP-A106_Lab03_C1126EHIT17343.zip: this report, the Evidence Appendix, the Evidence Manifest and Findings Matrix, the evidence log, screenshots with hash list, rebuilt text exports, README
Conventions	Each statement is labelled **Observed** (recorded by a tool), **Inferred** (my interpretation, with a confidence level) or **Not observed**. "Ex. En" points to an exhibit in the Evidence Appendix. Indicators are defanged. The malware family is not stated as a finding: the name of the archive is provenance only, and the behaviour is described as "consistent with" a file-encrypting ransomware.

1. Summary of findings

A packed 32-bit executable was triaged statically, followed through its unpacking stub in x32dbg to the original entry point (OEP), and its unpacked image was recovered from two memory dumps. The sample was also executed once without a debugger by mistake, which showed its real effect on the VM and is declared in section 11.

- **Packing (observed, high).** On disk the file has 8 sections, entry point RVA 0x3C40, only 6 imports in 4 libraries and file entropy 7.629. Sections .data, .crt and CODE are writable with entropy 7.65 to 7.88 (Ex. E01, E04, E03).
- **Unpacking (observed, high).** The stub ran from freshly allocated memory, called VirtualAlloc 3 times and VirtualProtect 6 times, restoring the final section protections of a 4-section image, and then control reached 004265E8 (OEP) in three separate runs (Ex. E54, E57, E52).
- **Memory (observed, high).** Two valid MDMP dumps (PID 4820 and PID 7152 at the OEP) contain the same .text, .rdata and .reloc sections (identical SHA-256 in both). The unpacked image is 4 sections with entry RVA 0x265E8.
- **Behaviour (observed).** Self-copy with the Hidden attribute, HKCU Run persistence, ransom-note files in many folders, user files renamed to .mp3 with an 8-byte header holding the victim ID, and a registry key HKCU\Software\xxxsys. Network use is a **capability only** (wininet, ws2_32, domains and a user-agent in memory); **no traffic was observed**.
- **Assessment (inferred, medium).** The behaviour is consistent with a file-encrypting ransomware. The victim ID, the Bitcoin-format address and the random self-copy names are absent from the unpacked image at the OEP and appear only in the post-run dump, so they are generated or written by the running payload.

Confidence. High for packing, the unpacking sequence, the OEP and the memory artefacts. High for file encryption, persistence and notes. Low for any claim about shadow-copy deletion (T1490) and for why the relaunched copies exited with code 1. The cipher, key handling and command-and-control behaviour were not analysed.

2. Environment and evidence preparation (Task A)

Virtual machine	FLARE-VM on VMware Workstation. Guest Windows 11 Pro 10.0.26200 (64-bit), host name DESKTOP-JV8FNEG, account `cybersky`, PowerShell 5.1.26100.7920. 8 GB memory, 4 processors, 102 GB NVMe disk.
Network	The adapter is Host-only with "Connected" and "Connect at power on" both unchecked (Ex. E09). The archive was brought in during a short window when the adapter was connected: connected at 11:18:57 with several flaps, last disconnect event 11:29:48 on 2026-10-03. The brief does not allow NAT or bridged networking, so this window is a declared deviation.
Shared folders and clipboard	Shared Folders: Disabled (a share definition exists but is disabled) (Ex. E64). Guest Isolation showed drag and drop and copy/paste enabled and greyed, as found (Ex. E63). This is a deviation from the brief. I did not use a shared folder or drag and drop. The console-text clipboard path was used to move text output (no files) from the VM to the report workspace.
Snapshots	Lab03_Clean_Baseline 2026-10-03 11:00:24 (taken before handling the archive); Lab03_PreDebug 11:41:26; Lab03_PostFirstRun 2026-10-04 01:20:21; Lab03_Suspended_4820 02:18:23; Lab03_PreOEP 11:42:33 (Ex. E11, E58, E59, E60, E61, E62). The clean snapshot is restored only after the package has been downloaded and confirmed.
Tools	PEStudio 9.61.0.0, Sysinternals Strings 2.54, x32dbg build "May 27 2026" (version 0.0.2.5, elevated, ScyllaHide 1.4.760), System Informer 3.2.25113.1114, 7-Zip 23.01, PowerShell 5.1 with a C# validator (Add-Type).
Working copy	The original is kept read-only in `original\sample\malware sample 4`. All analysis ran on `work\demo1_ransomware.exe` (identical hash, created 11:40:27). Hashes were recorded before debugging at 12:05:47.
Baseline	**Not recoverable.** The baseline PowerShell transcript from 11:02 on 2026-10-03 (111,012 B) was encrypted to `.txt.mp3` by the sample at 02:13 on 2026-10-04 (SHA-256 `99810D8A...F42A`). I did not decrypt it. After the final restore I will run the same baseline commands and record a post-restore verification.

3. Sample identification and hash chain

Item	SHA-256	Size	Note
Outer archive	`63E52C4A52CBFB9AB1885CBB83E5D5B836802A06618B3BC07EBC20545DF75729`		Provenance
Nested malware+sample+4.zip	`B049A5EC61205581C06C764A675983C26C2989374E22B167FD930DDC64E138CE`		Provenance
password (7).txt	`2F496C63E0BB77B76A65141982050E0D9CA887AE71B4856B4E984E0BDB4785F3`		Provenance
Source executable (O)	`5343947829609F69E84FE7E8172C38EE018EDE3C9898D4895275F596AC54320D`	368,640 B	MD5 9CE01DFB..., SHA-1 1BD767BE...
Working copy (W)	`5343947829609F69E84FE7E8172C38EE018EDE3C9898D4895275F596AC54320D`	368,640 B	Identical to O
Self-copies (5)	`5343947829609F69E84FE7E8172C38EE018EDE3C9898D4895275F596AC54320D`	368,640 B	gudpnujaukba, vltxuvwdbsbu, raknlbaoydys, wlcdcfqqewfn, bbflnrupddix (identical)

Item	SHA-256	Size	Note
Memory image (4820 dump)	42900DE68702ADB3E6802155AA416C7EEB5F19EBCC7D7387C08BAA4C51D7D9A8	544,768 B	Extracted at 0x400000
Memory image (OEP dump)	8DE53033FFB8C5936EE7BFA7BDFFA4C15ED1F2D98C9029693E4239C4A4EA75E5	544,768 B	Extracted at 0x400000
Dump PID 4820	6FA90986C74D9383B9FC0A9CC0AD3D06A92357E15EA95D4075AFEB4BFC79C0B8	117,739,615 B	Valid MDMP
Dump PID 7152 (OEP)	C93F7330B0EC11CF22915D246F56BC59F51DAFDE17BE724613531358C5F192AB	43,342,498 B	Valid MDMP
Dump PID 8532 (encrypted)	1F9F919EBC1D6077BE971762BAC200507D87EE45FA60FCBCF1D2B95D268669CC	132,232,656 B	Not a valid MDMP; encrypted by the sample

Observed: the source file was created 2021-04-23 06:11:30 with LastWriteTime 2017-04-01 18:11:02. The full chronological list of artefacts is in the Evidence Manifest. Raw dumps and executable images are not part of this submission.

4. Triage and packer assessment (Task B)

4.1 Static indicators (observed)

Property	Value
Format	32-bit GUI PE, image base 0x400000, entry point RVA 0x3C40, linker 8.0 (Visual Studio 2005)
Compile and debug stamp	Sun 2016-02-28 18:15:11 UTC (can be forged; not used as a finding)
Sections (8)	.text, para, .rdata, .data, .crt, CODE, .erloc, .rsrc
Entropy by section	.text 5.739; para 1.972; .rdata 1.030; .data 7.648; .crt 7.883; CODE 7.860. File entropy 7.629
Writable high-entropy sections	.data, .crt and CODE
Raw vs virtual size	.data raw 0xF000 (61,440 B) vs virtual 0x523D0; .crt raw 0x19000 (102,400 B) vs virtual 0x186B5. The in-memory sizes differ from disk
Imports	6 in 4 libraries: CLUSAPI.dll GetClusterResourceKey; msvcrt memset, memcpy; KERNEL32 GlobalMemoryStatus (flagged), CreateEventW; USER32 RemovePropA. imphash C00702BDB5E1419C3DC899A74A60A37D
Resources	13 entries, 34,403 bytes; one PNG-containing icon entry with entropy 7.892
Debug directory	E:\Tools\aolfed\release\osc.pdb, RSDS GUID 2FD65FFB-5681-4310-835F-ED44E8CFD90
Version info	"nah nah Corporation", "nah nahApp", 1.600.5512 (descriptive text, not evidence of origin)
Strings	PEStudio counted 10,696 strings. Strings -n 8 found 162 ASCII and 34 Unicode, including VirtualAlloc, GetClusterResourceKey, CLUSAPI.dll, RemovePropA, "pewpewpew", "die hard", "Joyhv.pew", "nbvgg.hlp", "control.ini"

Exhibits: Ex. E01, E02, E03, E04, E05, E06, E07, E08.

4.2 Assessment

Inferred, high confidence: packed. Four independent signals agree: an import table of only 6 functions and no file, registry or network API; three writable sections at entropy 7.65 to 7.88 whose virtual size far exceeds the raw size; a very small entry stub; and no readable indicators on disk (the 4 payload domains, the onion address and the note-file names are absent from the file). The packer was not identified by name. The statement is limited to "a custom or unidentified packer" because no packer was named by the tools used.

Why it matters. Static output on the packed file describes the stub, not the payload. The 6 imports (VirtualAlloc is resolved at run time, per the debugger) cannot be used to judge capability. This is why the analysis moved to the debugger and to memory.

5. Controlled unpacking and entry point analysis (Task C)

5.1 Method

- x32dbg (elevated, ScyllaHide loaded) on the working copy in the VM; the sample was never run outside the debugger by intent.
- Breakpoints on the entry stub (00403C40), `kernel32.VirtualAlloc` (751565F0), `kernel32.VirtualProtect` (75156B30), CreateFileW, CreateProcessW and ExitProcess, with log-and-continue so the arguments of each call were recorded.
- Hardware execute breakpoints (`bphws`, 4 slots available) on 00403C40, 0041EAA0, 0041FEB0 and 004265E8. The OEP address was chosen from the pattern of VirtualProtect calls and then confirmed by the hit.

5.2 Debug sessions

PID	Time	Session	What it shows
7380	10/3 12:12	First debug	System breakpoint; no run past the stub
8360	10/3 12:28	TLS stop	TLS callback stop. System Informer later shows the parent of PID 8532 as non-existent process 8360
7784	10/3 12:45 to 12:55	TLS and entry stops	Entry stop at 00403C40. The wmic dialog was on screen (originating process not attributed)
3848	10/4 01:16	Log	System breakpoint log
5348 (S1)	10/4 11:46 to 11:53	Log, exit code 1	Inferred as S1 from matching Log lines
not captured (S2)	10/4	Log, exit code 1	PID not captured
6068 (S3)	10/4 12:08 to about 12:43	Entry stop, VirtualAlloc, VirtualProtect	Exit code 1
7152 (S4)	10/4 12:43	OEP hit	Exit code 0, dump taken at 12:50:14

Observed: the x32dbg Log tab accumulates across sessions and shows four session boundaries: S1 lines 44 to 160, S2 165 to 282, S3 287 to 411, S4 416 to 499. S1 to S3 ended with exit code 1; S4 ended with exit code 0.

5.3 Stub behaviour (observed)

At the entry stop `00403C40` the instructions save the registers to fixed data addresses (`mov dword ptr ds:[414DA0],esi; push ebp; ... mov [414DA8],ebx`) and then `jmp 4013D0`. The VirtualAlloc breakpoint was hit 3 times and the six VirtualProtect calls below followed. The return addresses (026Bxxxx in S3, 024xxxxx in S4) are in freshly allocated memory, so the stub that changes the protections is not part of the on-disk image.

#	Address	Size	New protection	Region	Return address / exhibit
1	`0x400000`	`0x9E000`	0x4 PAGE_READWRITE	whole image	024425D4 (S4); 026B25D4 (S3); Ex. E47
2	`0x400000`	`0x400`	0x2 PAGE_READONLY	PE headers	02442670 (S4); 026B2670 (S3); Ex. E48
3	`0x401000`	`0x2F3D0`	0x20 PAGE_EXECUTE_READ	.text	024427FF; Ex. E49

#	Address	Size	New protection	Region	Return address / exhibit
4	0x431000	0xA9C0	0x2 PAGE_READONLY	.rdata	024427FF; Ex. E50
5	0x43C000	0x45304	0x4 PAGE_READWRITE	.data	024427FF; Ex. E51
6	0x482000	0x2BD8	0x2 PAGE_READONLY	.reloc	024427FF; Ex. E57

Calls 1 to 2 make the whole image writable and then restore the headers to read-only; calls 3 to 6 set the final section protections of the unpacked image. The protections match the unpacked section table (.text executable, .rdata read-only, .data read-write, .reloc read-only). Calls 2 to 4 recorded in S3 (PID 6068, 12:21 to 12:39) match the same calls in S4 (PID 7152). Call 6 and its arguments come from the S4 Log screenshot (Ex. E57).

5.4 Original entry point (observed)

The hardware execute breakpoint at `004265E8` was hit (Ex. E52). The hit count is 1 (Ex. E54). Disassembly at the OEP starts `call 0x42B9D5; jmp 0x42647B; ... push ebp; mov ebp,esp; sub esp,328`, the pattern of a Microsoft C runtime start-up. The OEP is consistent with the entry RVA 0x265E8 in the unpacked header (0x4265E8 minus the base 0x400000). The OEP was reached in three separate runs. After the OEP, a thread was created (6912), the process ended with exit code 0 and the debugger stopped.

Function prologues (55 8B EC) in the OEP dump at RVA 0x1EAA0, 0x1FEB0 and 0x13820 match the thread start addresses of the running copy 8532 (gudpnujaukba.exe+0x1eaa0, +0x1feb0, +0x13820, +0x3c40). The breakpoints at 0041EAA0 and 0041FEB0 never fired in debugged runs (0 hits). I did not determine why. This is an open question, not a finding.

5.5 Behaviour after the OEP in S1 to S3 (observed)

After the OEP the log shows CreateFileW twice, CreateProcessW, loading of urlmon and winhttp, CreateProcessW again, ExitProcess and exit code 1. The arguments of CreateFileW and CreateProcessW were not read, so the targets are not stated. A self-copy was written to `C:\Windows` and an HKCU Run value was created in each of S1 to S3; Prefetch shows each copy running 2 to 4 seconds later. **Hypothesis (not tested):** a single-instance mutex or another environment check causes the exit code 1.

6. Memory acquisition and artefact validation (Task D)

6.1 Acquisition

Artefact	Size	Process state	Validity
gudpnujaukba_8532.dmp.mp3	132,232,656 B	PID 8532 (uncontrolled first run), dump started 01:36	Not valid. The file was encrypted by the sample after creation; header bytes 00 x8, then 48 55 EB 75 F8 8F BD 05. Excluded from analysis
bbflnrupddix_4820.dmp	117,739,615 B	PID 4820 started 01:50:03, suspended, 7 threads, parent shown as non-existent process 1120	Valid MDMP, 357 regions, 55 modules, stream types 3,17,4,14,9,16,7,15,12,21,22,0, main image 0x400000 size 0x85000
demo1_ransomware_7152_oep.dmp	43,342,498 B	PID 7152 stopped at OEP 004265E8, 12:50:14	Valid MDMP, 218 regions, 30 modules, stream types 3,17,4,9,16,7,15,12,21,22,0

Dumps were created with System Informer (Create dump file) and validated with a C# parser written during the exercise because the system Python was unusable (its standard library files had been encrypted). The check covers the MDMP signature, stream types, region and module counts, the PE header of the image at 0x400000 and the per-section SHA-256.

6.2 Unpacked image (observed)

Section	VA	Virtual size	Entropy (memory)	SHA-256 (both dumps unless noted)
.text	0x1000	0x2F3D0	6.69	2A84B69358856CC835F476E4CF1FCEA5C3E5E1641C210C4763CB649EBF543815
.rdata	0x31000	0xA9C0	6.18	0FBAF5E7EE4A714902D29A4C79A88B8015FABE01E268DA31B0C8C28BAB2910B7
.data	0x3C000	0x45304	0.30 (OEP) / 0.94 (4820)	OEP 6267F86336983B49B960EE133CC4E5E35C07BF67EA882D92F88434A175D944CD; 4820 DF9934A738ABB8C4428E7983D4B8B85C9F346D4A40071078741264C2AAD1E5A1 (differs: runtime data)
.reloc	0x82000	0x2BD8	5.11	278E76127ACDCDDF7DAA5FFD9E008C9276A4724B6D703F20797EB93B1165713A

Observed: SizeOfImage 0x85000, TimeDateStamp 0x56D208AF (2016-02-27 20:35:59 UTC), EntryRVA 0x265E8. The first 0x400 header bytes differ from the file on disk. The three code and constant sections are identical in both dumps, which supports a stable unpacked image; .data differs because it holds runtime state. The columns "file entropy" and "differing" in the validator output compare against a mismatched section table and are **not cited**.

6.3 Indicators in the unpacked image compared with disk (observed)

String	OEP dump	4820 dump	Disk	Note	First VA (OEP)
zatcurr (domain)	6	18 (3 domains)	0		0x438C15
keratadze, milerteddy	6 each	in the 18	0		0x438C46 / 0x438C78
.onion	3	9	0		0x438D95
Victim ID 4855EB75F88FBD5	0	23 ASCII, 1 UTF-16	0	post-run only	4820: 0x460C85
Bitcoin-format address	0	1	0	post-run only	4820: 0x43FC58
RECOVERY	1 UTF-16	666 UTF-16	0		
wmic	2 ASCII, 1 UTF-16	3 ASCII, 14 UTF-16	0		0x43A2A8
vssadmin	0	0	0	Not found	
CurrentVersion\Run	2 UTF-16	3 UTF-16	0		
.mp3	4 UTF-16	30 UTF-16	0		
gudpnujaukba (OEP) / bbflnrupddix	0 / 0	- / 1 ASCII, 36 UTF-16	0	random name, post-run	4820: 0x729754

Inferred, medium confidence. The three domains, the onion address and the note-file marker exist in the unpacked image at the OEP, so they are configuration of the payload. The victim ID, the Bitcoin-format address and the random self-copy names are absent at the OEP and appear only after the payload runs, so they are generated or written at run time. This is why a dump taken only at the OEP is not enough to extract them.

7. Runtime behaviour (observed unless stated)

7.1 Uncontrolled first run, PID 8532

The sample was started at 12:45:25 on 2026-10-03 from `C:\Windows\gudpnujaukba.exe` with the working directory `C:\Labs\CIP-A106-Lab03\work\`. At 01:30 to 01:37 on 10-04, System Informer showed about 40% CPU, I/O of 1.4 to 1.7 MB/s, 4 sample threads (+0x1feb0, +0x13820, +0x3c40, +0x1eaa0) and cumulative disk I/O of 11.22 GB read and 5.72 GB written (Ex. E22, E27, E28). Loaded modules include clusapi, resutils, mi and miutils (Ex. E23). The dump of this process was encrypted (section 6.1).

7.2 File system

- **Self-copies.** Five 368,640-byte copies with the same SHA-256 as the source: `C:\Windows\gudpnujaukba.exe`, `vltxuvwdbsbu.exe`, `raknlbaoydys.exe`, `wlcdcfqqewfn.exe` and `Documents\bbflnrupddix.exe`. The three Windows copies carry Hidden+Archive and a preserved LastWriteTime of 2017-04-01 18:11:02.
- **Notes.** `_RECOVERY_+<suffix>.txt/.html/.png` (1,953 B / 11,695 B / 60,935 B for the suffixes ylvad and xshgj, identical hashes), `recover_file_*.txt` (253 B) and `RECOVERY.TXT/HTM/png` on the Desktop, in many folders including the Startup folders. Note text describes AES encryption, a "Secret Server", Bitcoin payment, three personal-page URLs and a Tor address. I did not follow any of them.
- **Encrypted files.** User files renamed to `.mp3` (8,009 counted at 02:09). Header: 8 zero bytes, then the victim ID bytes `48 55 EB 75 F8 8F BD 05`. Evidence files, PowerShell transcripts and Python standard library files in the VM were also encrypted.
- **Decoys.** Five 14-byte decoy `.txt` files placed before the run were not encrypted: decoy1 1C921752, decoy2 EDAC0803, decoy3 3B588043, decoy4 FDC19E65, decoy5 DD72CEFD. I did not determine why (see limitations).

7.3 Registry

- HKCU Run values at 13:30: five malicious values (iorpyhcbjsja to gudpnujaukba.exe, vylvawuyxpdy to Documents\bbflnrupddix.exe, lccjvxmpqmjl to vltxuvwdbsbu.exe, objdqnybfhvp to raknlbaoydys.exe, wikqxbughfcu to wlcdcfqqewfn.exe), each `cmd.exe /c start "" "<copy>"`. HKLM Run was clean (SecurityHealth, VMware User Process, BinDiffPerUserSetup). The five values were removed at 13:32 and the key was exported (`15_hkcu_run_after.reg`).
- `HKCU\Software\xxxsys\ID` = 4855EB75F88FBD05. `EnableLinkedConnections` = 1 was found; its origin was not verified, so it is not attributed to the sample.

7.4 Processes and the wmic dialog

The Windows dialog "Windows cannot find 'wmic.exe'" appears in screenshots at 12:52 and 12:55 on 10/3 and 1:33 and 1:55 on 10/4 (Ex. E18, E29). The originating process was not attributed in the screenshots, and WMIC is absent on this Windows build. **Inferred, low confidence:** an attempt to run a wmic command, consistent with the string "wmic" in the payload. No shadow-copy deletion effect was observed or tested.

7.5 Network

Capability only. The unpacked image imports wininet, ws2_32, iphlpapi, mpr, ndfapi and wdi, and contains a Trident user-agent string, three domains and a Tor address. The adapter was disconnected and no packet capture was run, so **no traffic was observed**. I make no claim about command-and-control behaviour.

8. Indicators of compromise

Type	Value (defanged)	Where found	Confidence
Victim ID	`4855EB75F88FBD5 (registry and file-header form 4855EB75F88FBD05)`	Ransom note text; HKCU\Software\xxxsys\ID; first 16 bytes of encrypted files	High
Bitcoin-format address	`1E3Hf3Mpy4g8zZUHPoAFwSarq5UxLbocT3`	Ransom note; payload memory after the run (format only, not validated on any ledger)	High (presence)
Domain	`gwe32fdr74bhfsyujb34gfszfv[.]zatcurr[.]com`	Unpacked image (6 hits) and notes; URL form hxxp://<domain>/<ID>	High (presence); no traffic observed
Domain	`tes543berda73i48fsdfsd[.]keratadze[.]at`	As above	High (presence); no traffic observed
Domain	`tt54rfdjhb34rfbnknaerg[.]milerteddy[.]com`	As above	High (presence); no traffic observed
Onion address	`xlowfznrg4wf7dli[.]onion/<ID>`	Unpacked image (3 hits) and notes	High (presence); no traffic observed
Registry key	`HKCU\Software\xxxsys (value ID)`	Registry query after first run	High
Run values (HKCU)	`iorpyhcbjsja, vylvawuyxpdy, lccjvxmpqmjl, objdqnybfhvp, wikqxbughfcu, each cmd.exe /c start "" "<copy>"`	Get-ItemProperty 2026-10-04 13:30	High
File name (self-copies)	`C:\Windows\gudpnujaukba.exe, vltxuvwdbsbu.exe, raknlbaoydys.exe, wlcdcfqqewfn.exe; Documents\bbflnrupddix.exe`	File listings; identical SHA-256 to the source	High
File name (notes)	`_RECOVERY_+<suffix>.txt/.html/.png, recover_file_<suffix>.txt, RECOVERY.TXT/HTM/png`	File listings	High
File extension	`.mp3 appended to user files; first 8 bytes zero, then victim ID`	Header bytes of encrypted files; 8,009 .mp3 files counted at 02:09 AM	High
Hash (SHA-256)	`5343947829609F69E84FE7E8172C38EE018EDE3C9898D4895275F596AC54320D`	Source sample and all five self-copies	High
Hash (imphash)	`C00702BDB5E1419C3DC899A74A60A37D`	PEStudio	High
PDB path	`E:\Tools\aolfed\release\osc.pdb`	PEStudio debug directory	High
Version strings	`"nah nah Corporation", "nah nahApp", 1.600.5512`	PEStudio	High (provenance only)
Strings	`pewpewpew, die hard, Joyhv.pew, nbvgg.hlp, control.ini, kasdfgh283, ityeofm9234-23423 (purpose unknown)`	Strings v2.54 and unpacked image	Medium
User-agent	`Mozilla/5.0 (Windows NT 6.3; WOW64; Trident/7.0; Touch; rv:11.0) like Gecko`	Unpacked image	High (presence); no traffic observed

9. MITRE ATT&CK mapping

ID	Technique	Status	Confidence	Basis
T1027.002	Obfuscated Files or Information: Software Packing	Observed	High	Section entropy 7.86 to 7.88 (.crt, CODE), 6 imports, entry in a small stub, in-memory image differs from disk, six VirtualProtect calls then OEP hit (Ex. E01, E04, E57, E52)
T1486	Data Encrypted for Impact	Observed	High	User files renamed to .mp3 with an 8-byte header holding the victim ID; notes dropped; evidence and transcripts encrypted. The encryption algorithm was not examined (Ex. E29, E28)
T1547.001	Boot or Logon Autostart: Registry Run Keys	Observed	High	Five HKCU Run values, each cmd.exe /c start "" "<copy>" (console output 13:30)
T1564.001	Hide Artifacts: Hidden Files and Directories	Observed	High	Self-copies carry Hidden+Archive attributes
T1036	Masquerading	Inferred	Low	Random self-copy names and the .mp3 extension on encrypted files. No attempt to imitate a legitimate program name was seen
T1112	Modify Registry	Observed (xxxsys); unverified (EnableLinkedConnections)	Medium	HKCU\Software\xxxsys\ID written. The origin of EnableLinkedConnections=1 was not verified
T1490	Inhibit System Recovery	Attempted, not confirmed	Low	A "Windows cannot find wmic.exe" dialog appeared in four screenshots and the string wmic is in the payload image (3 hits). The command line was not captured and WMIC is absent on this Windows build, so no shadow-copy deletion effect is claimed (Ex. E18, E29)

Techniques not mapped because they were not observed: command and control over the network, credential access, process injection, lateral movement.

10. Findings matrix

The findings matrix links each IOC, behavioural conclusion and ATT&CK mapping to its source artefact, confidence level and limitation. It is a separate deliverable (Evidence Manifest and Findings Matrix, section 2, and `findings_matrix.csv`). The key rows are repeated here.

Finding	Source artefact	Confidence	Limitation
Packed with an unidentified packer	PEStudio views; entropy; 6 imports (E01-E08)	High	Packer not named
Stub allocates memory and restores section protections	x32dbg Log (E57), breakpoint hits (E54)	High	Call 6 arguments from S4 only
OEP at 004265E8 (RVA 0x265E8)	E52, E54; dumps; header	High	None for the three runs
Unpacked image recovered from memory	Two valid MDMP dumps; section hashes	High	.data differs (runtime state)
Victim ID and Bitcoin address appear only after the payload runs	String tables, OEP vs 4820 dump	Medium	Two dumps only; absence is not proof
File encryption (T1486)	File headers, .mp3 files, notes, E29	High	Cipher and key handling not analysed
Run key persistence (T1547.001)	Console output 13:30; reg export	High	Three of five values created during debug sessions S1 to S3

Finding	Source artefact	Confidence	Limitation
Shadow-copy deletion attempt (T1490)	wmic dialog (E18, E29); wmic string	Low	Command line not captured; WMIC absent
Network capability	Imports, strings in memory	Medium	No traffic captured

11. Deviations and limitations

- **Network window.** The adapter was connected from about 11:18:57 to 11:29:48 on 2026-10-03 to bring in the archive. NAT or bridged networking is not allowed by the brief. No sample was present then.
- **Guest Isolation.** Drag and drop and copy/paste show as enabled. The shared folder is defined but disabled. Only console text was moved by clipboard; no files were moved.
- **Uncontrolled execution.** At 12:45:25 the sample ran outside the debugger. It wrote self-copies, a Run value, notes and encrypted user, evidence, earlier-lab, transcript and Python files in the VM. The brief forbids letting the sample touch real user files; the VM held only lab data, but this is a procedural failure and is declared. The VM was rebooted at 01:44:43 on 10-04.
- **Baseline lost.** The 11:02 baseline transcript was encrypted (section 2). A post-restore verification will substitute and is not a pre-execution record.
- **Extra persistence artefacts.** Sessions S1 to S3 created three more self-copies and Run values, removed at 13:32. PID 4820 and the five copies remain in the VM until the restore.
- **Evidence gaps.** PID of S2 and the parent of PID 7152 not captured; CreateFileW and CreateProcessW arguments not read; OEP hardware hit not logged in S1; the full 26 KB x32dbg log is reproduced only as excerpts; EnableLinkedConnections origin unverified; reason for the exit code 1 in debugged runs unknown; purpose of the threads at 0041EAA0 and 0041FEB0 unknown.
- **Rebuilt text files.** The text exports in the package are rebuilt from console output pasted out of the VM. Their line endings differ from the originals, so their hashes do not match; the hash of each VM original is given in the Evidence Manifest.
- **Validator columns.** The "file entropy" and "differing" columns of the memory validator used mismatched section tables and are not cited.
- **No packet capture, no cipher analysis, no family attribution.**

12. Conclusion

The sample is a packed 32-bit executable whose stub allocates memory, rewrites section protections in six VirtualProtect calls and transfers control to an original entry point at 004265E8. The unpacked image, recovered from two valid dumps, contains the payload configuration (three domains, an onion address, note-file markers) and the code for the behaviour observed: self-copy with the Hidden attribute, Run-key persistence, ransom notes and file encryption with a victim ID in the header. This is consistent with a file-encrypting ransomware. The cipher, the network behaviour and the shadow-copy attempt were not confirmed. The most important process lesson is that the sample ran once outside the debugger; with that declared, the evidence still supports Tasks A to E with the limits stated above.